MATH3030 Coursework

Multivariate Statistics — Spring 2026

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1 Question 1

1.1 Exploratory Data Analysis

The UN dataset contains GDP per capita, life expectancy, and population for 141 countries at 12 evenly spaced time points from 1952 to 2007. No values are missing. Sample sizes per continent are uneven (52 African, 32 Asian, 30 European, 25 American, and 2 Oceanian), so any continent-level summary for Oceania is effectively a two-country average of Australia and New Zealand. We summarise the 70-year dynamics with continent-mean trajectories for each variable and a table of 1952 and 2007 medians.

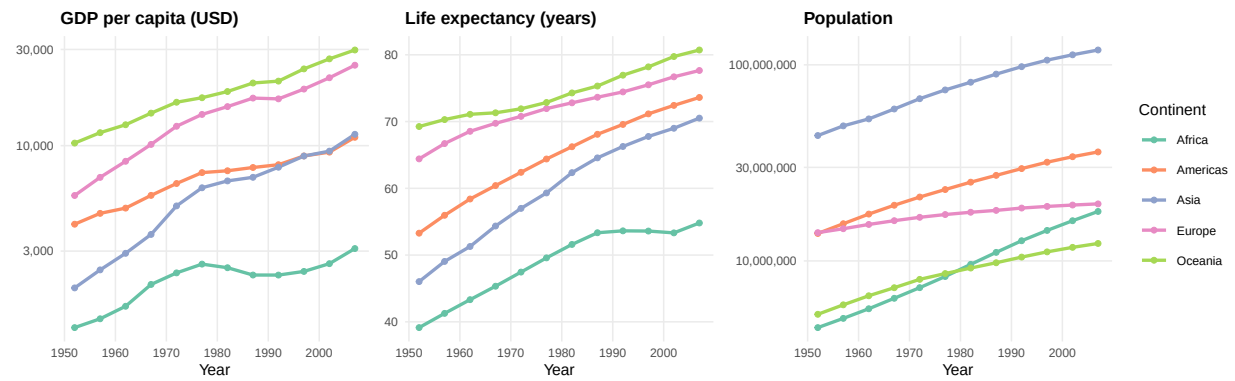

Figure 1: Continent-mean trajectories, 1952–2007. GDP and population are on log scales; life expectancy is linear.

Asia gains most on both axes. Mean life expectancy rose from 44 to 72 years and median GDP per capita almost quadrupled. The continent mean hides a wide spread. Singapore, South Korea, and Taiwan each grew GDP per capita by 20–24×, far above the 3.8× Asian median. China alone accounts for the population panel, rising from 556 million in 1952 to 1.32 billion in 2007.

Africa's trajectory is much shallower. Median GDP rose only 1.5× over 55 years, and mean life expectancy, after rising from 39 to 53 years, plateaued from the 1990s onward. The plateau coincides with a broad health and conflict downturn in southern and central Africa. Zimbabwe's life expectancy fell from 48.5 to 43.5 years (the largest decline in the dataset), Swaziland, Zambia, and Lesotho show similar stagnation, and the Democratic Republic of the Congo is the only country whose GDP per capita ended below its 1952 level, at 36% of its original value. Africa's population grew faster than any other continent's, which compounds the GDP-growth shortfall on a per-capita basis.

Europe and Oceania remain at the top throughout. Europe's GDP growth ($5,143 to $28,054, 5.5×) matches Asia's in multiplier terms, but its life-expectancy gain of +13 years is about half of Asia's +28. This is a ceiling effect. A population already close to the demographic ceiling has less room to gain years than one catching up. Both continents evolve more smoothly than Asia or Africa, with no sharp deviations between adjacent five-year intervals.

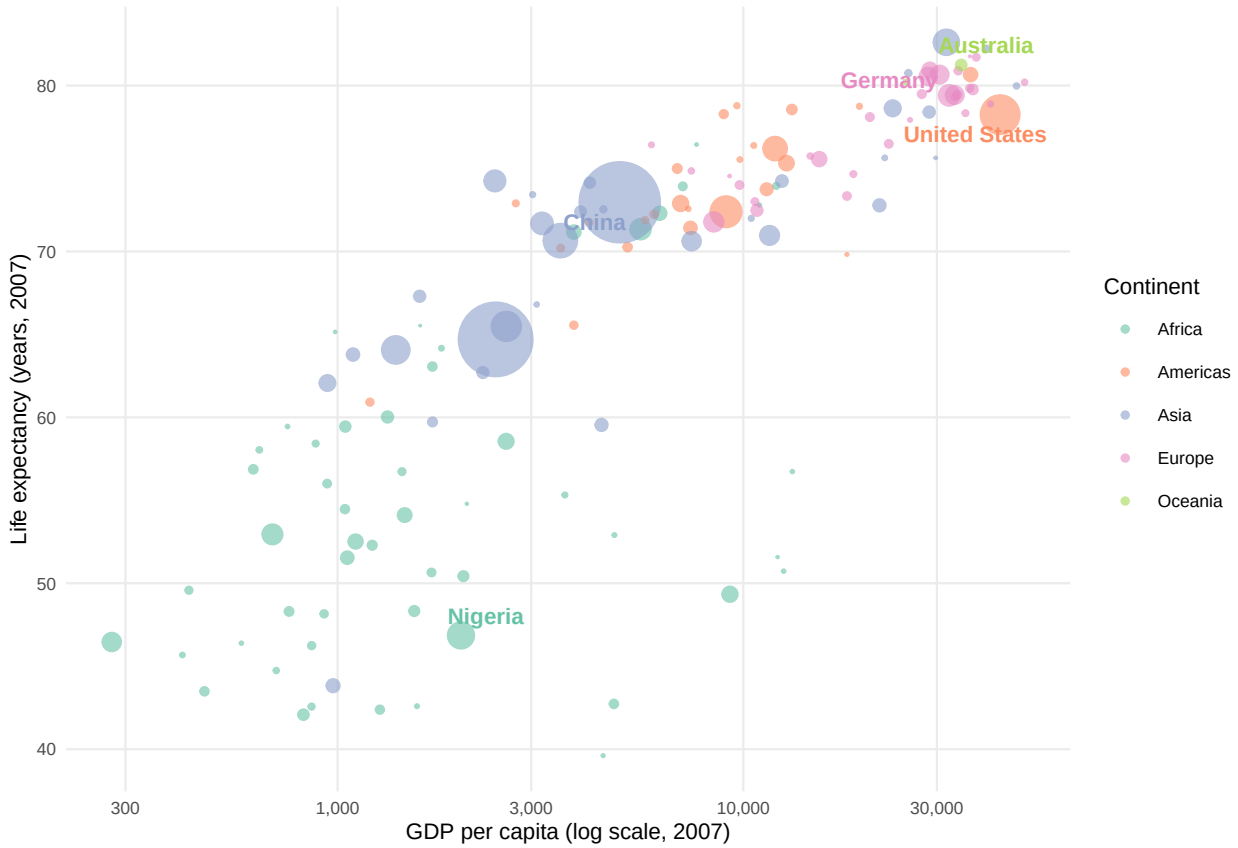

Figure 2: GDP per capita vs life expectancy in 2007. Bubble size is proportional to population.

GDP and life expectancy correlate at 0.81 on the log scale and 0.68 on the raw, the gap reflecting diminishing returns on life expectancy at high income. Asia is the only continent spanning the full income range, from India and Bangladesh up to Japan and Singapore, with China between the two modes.

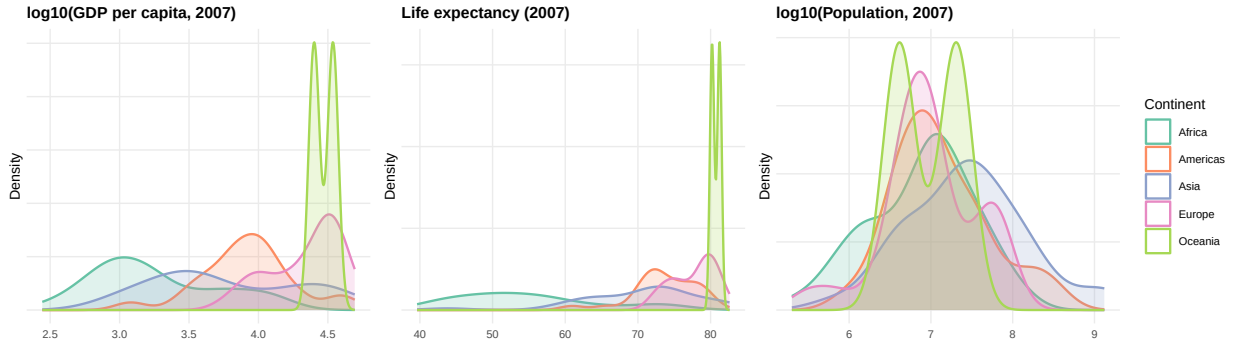

Figure 3: Kernel density estimates of each variable in 2007.

Asia's GDP distribution is bimodal, splitting Gulf states and East Asian tigers from South and Central Asia, with an IQR ratio of 8.9, the largest of any continent. Africa's life-expectancy

distribution is wide and left-skewed, with a sub-45-year tail in southern African countries. Asia carries the long population right tail (China, India). The continent means hide all of this.

Table 1: Median GDP per capita (USD) and life expectancy (years) by continent; population (millions).

	N	GDP 1952	GDP 2007	LE 1952	LE 2007	Pop 2007 (M)
Africa	52	987	1452	38.8	52.9	930
Americas	25	3048	8948	54.7	72.9	899
Asia	32	1148	4328	44.4	72.2	3809
Europe	30	5142	28054	65.9	78.6	586
Oceania	2	10298	29810	69.3	80.7	25

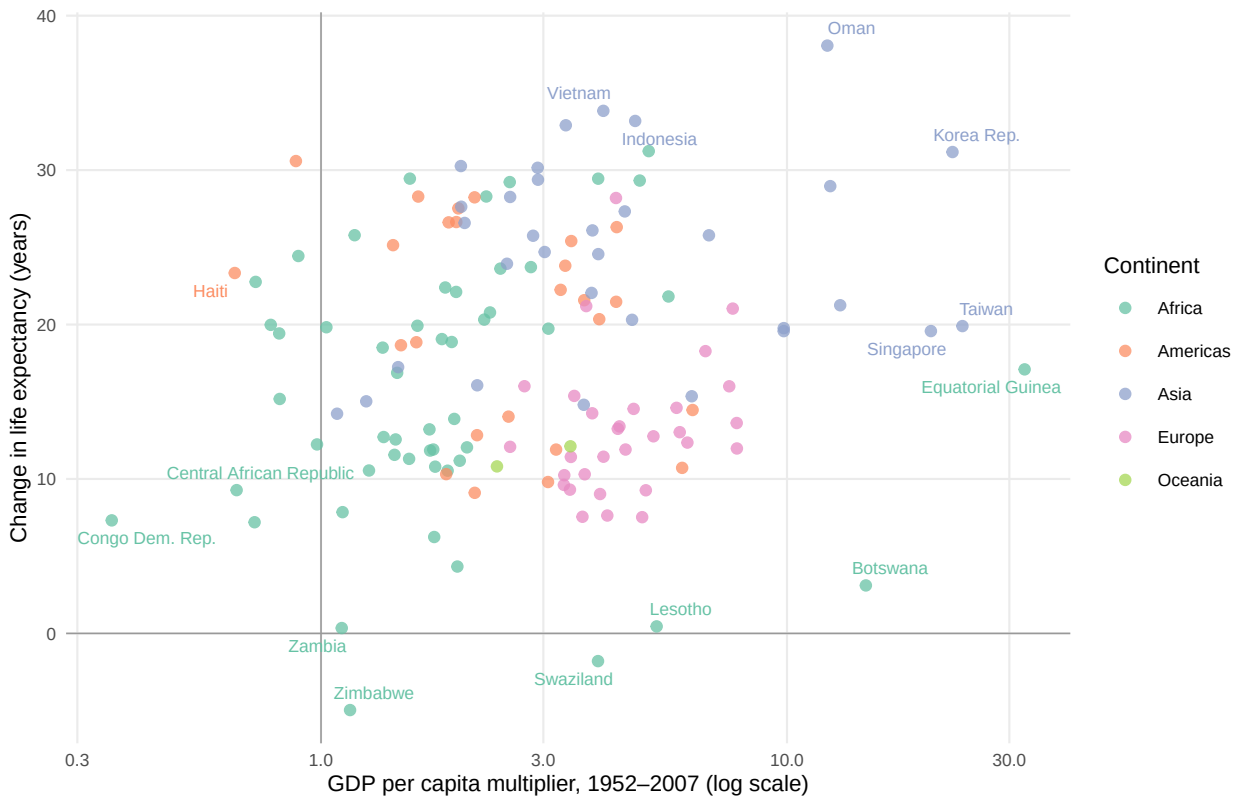

Figure 4: GDP-per-capita multiplier (1952→2007, log scale) against absolute change in life expectancy. Labelled points are the largest deviations on either axis.

A small number of countries deviate strongly from their continent's trend and reappear as extreme points in the PCA, CCA, and MDS projections below. Equatorial Guinea's GDP per capita rose 32× on the back of offshore oil discoveries, Zimbabwe's life expectancy fell by five years, Singapore, South Korea, and Taiwan achieved sustained 20–24× GDP multiples, and the Democratic Republic of the Congo and Haiti are the two economies that regressed most sharply, ending the period at 36% and 65% of their 1952 GDP per capita respectively. The two axes are largely decoupled, with top GDP multipliers and top life-expectancy gains landing on different countries.

1.2 Principal Component Analysis

We perform PCA independently on the 141×12 GDP and life-expectancy blocks. GDP values span more than two orders of magnitude across countries, so we standardise each year column (which is equivalent to decomposing the correlation matrix) to prevent larger recent-year variances from dominating. Life expectancy is on a comparable scale across years, but we also standardise it. The covariance and correlation decompositions produce near-identical results (PC1 explains 92.0% and 92.3% of the variance respectively), and standardising keeps the treatment of the two variable blocks consistent.

1.2.1 Scree analysis and variance explained

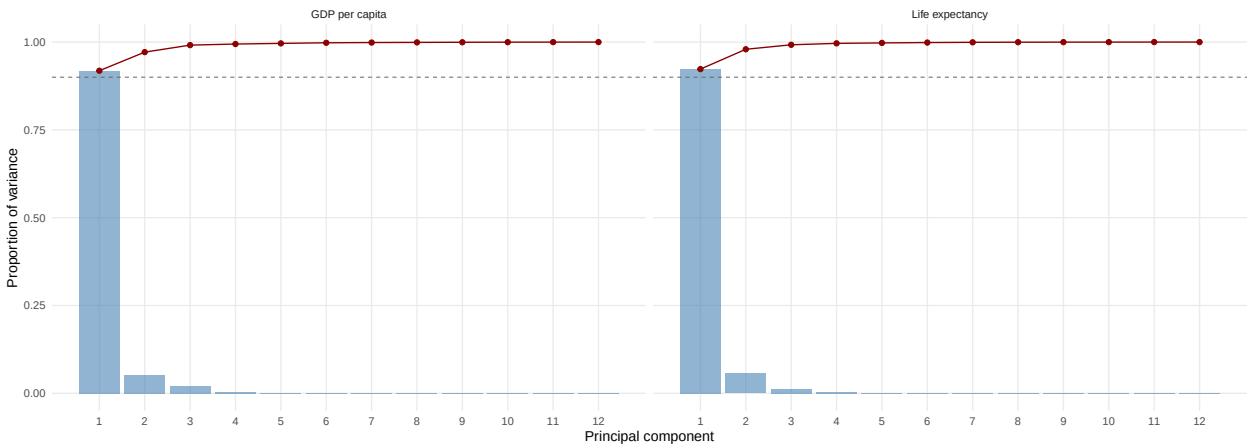

Figure 5: Scree plots for the GDP and life-expectancy PCAs.

Both decompositions concentrate almost all variance in a single component. PC1 explains 91.8% of the GDP variance and 92.3% of the life-expectancy variance while PC2 adds a further 5.3% and 5.6%. Two components cover 97.1% of GDP and 97.9% of life expectancy. Adjacent years correlate at ≈ 0.99, so 12 columns compress to two.

1.2.2 Interpretation of PC1 and PC2

The loadings have the same structure on both decompositions. PC1 assigns nearly uniform positive weights across all 12 years (0.28 to 0.30 on GDP, 0.27 to 0.30 on life expectancy), so a country's PC1 score is approximately proportional to its average level over the full 55-year period. For PC2, the loadings begin strongly negative in 1952 (-0.35) and rise monotonically to strongly positive in 2007 ($+0.42$). A positive PC2 score therefore indicates a country whose values were low early and high late (a catching-up or fast-growth trajectory), and a negative PC2 score indicates the opposite (stagnation or decline from a relatively high base).

1.2.3 PC score scatter plots

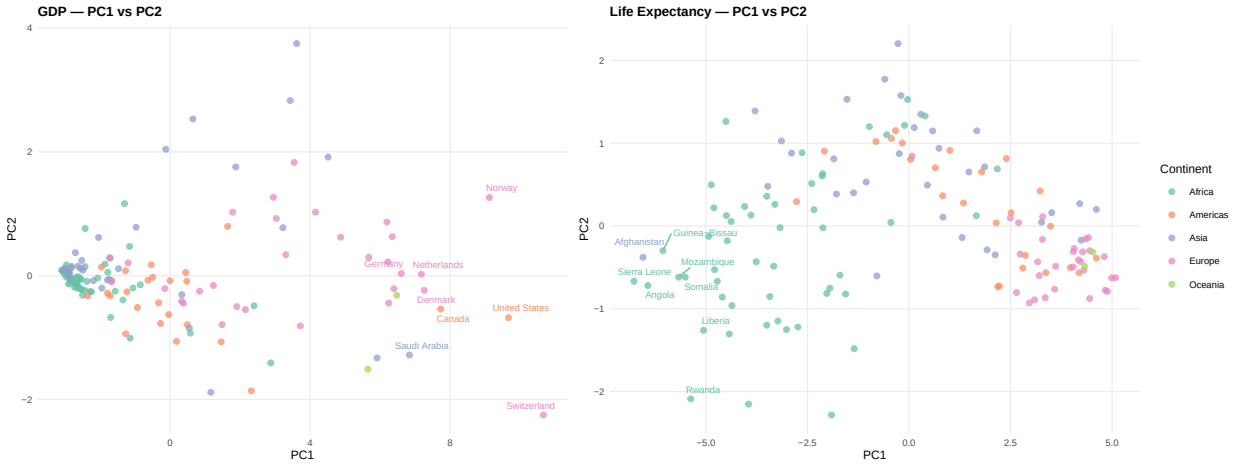

Figure 6: PC1 vs PC2 scatter plots for GDP (left) and life expectancy (right), coloured by continent.

On the GDP scatter PC1 ranks countries by income level (Switzerland and the US at the top, the poorest African economies at the bottom). PC2 then picks out growth trajectory, with the East Asian tigers clustering at the top, while Venezuela, Iraq, and Libya sit at the negative end as 1952 high-income economies that lost ground.

The life-expectancy scatter has the same shape. PC1 again ranks by level (Scandinavian and Alpine Europe at the top, sub-Saharan Africa at the bottom) and PC2 by trajectory, with Oman and Vietnam at one end and Zimbabwe and Botswana at the other, matching the southern African downturn from the EDA. Asia stretches along PC2 because of its bimodal income distribution; Europe and Oceania sit together at the top of PC1, Africa at the bottom, the Americas between.

1.2.4 GDP PC1 versus life-expectancy PC1

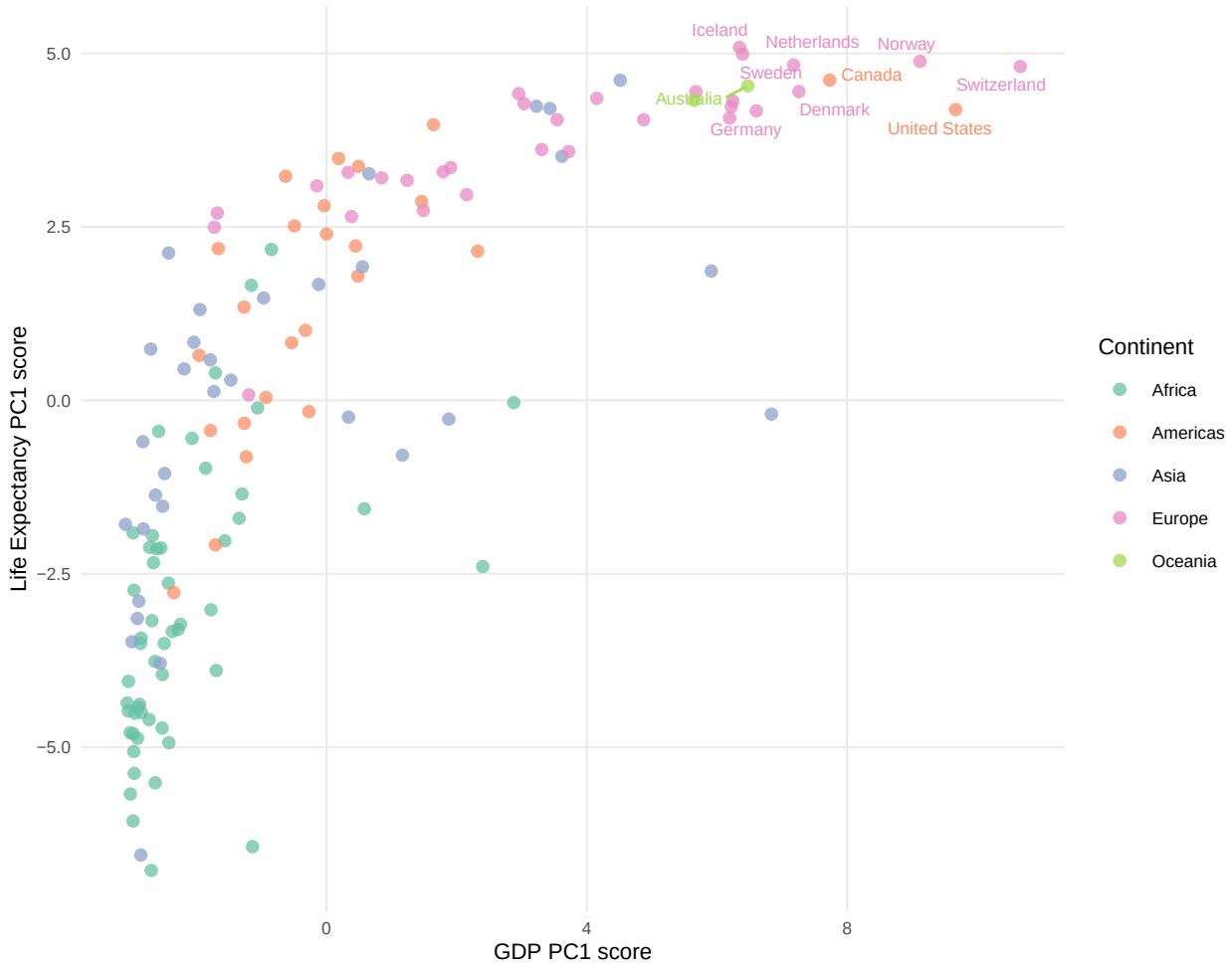

Figure 7: First PC scores, GDP against life expectancy.

The two PC1 scores correlate at 0.77, in line with the 2007 cross-sectional 0.81 on log(GDP). Most countries lie close to the diagonal; deviations are countries whose health outcomes are stronger or weaker than income alone would predict, picking up public-health investment, diet, and health-system access.

1.2.5 Limitations of linear PCA

The decomposition is restricted to linear combinations of the year columns and the GDP–life expectancy relationship is non-linear (Pearson correlation 0.68 on raw GDP against 0.81 on log(GDP)). Kernel PCA, which applies the standard decomposition in a feature space induced by a kernel, would capture this curvature directly and would be expected to produce tighter continent separation than the linear projection used here.[1]

[1] Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed., 2009), Chapter 14.

1.3 Canonical Correlation Analysis

We apply CCA with log(GDP) as the first variable set and life expectancy as the second, giving $p = q = 12$. The log transform is motivated by the same skewness and non-linearity noted in the EDA and the PCA limitations subsection. Raw GDP is strongly right-skewed (skewness 1.88) and the 2007 GDP–life expectancy relationship curves noticeably on the raw scale but is close to linear on the log scale.

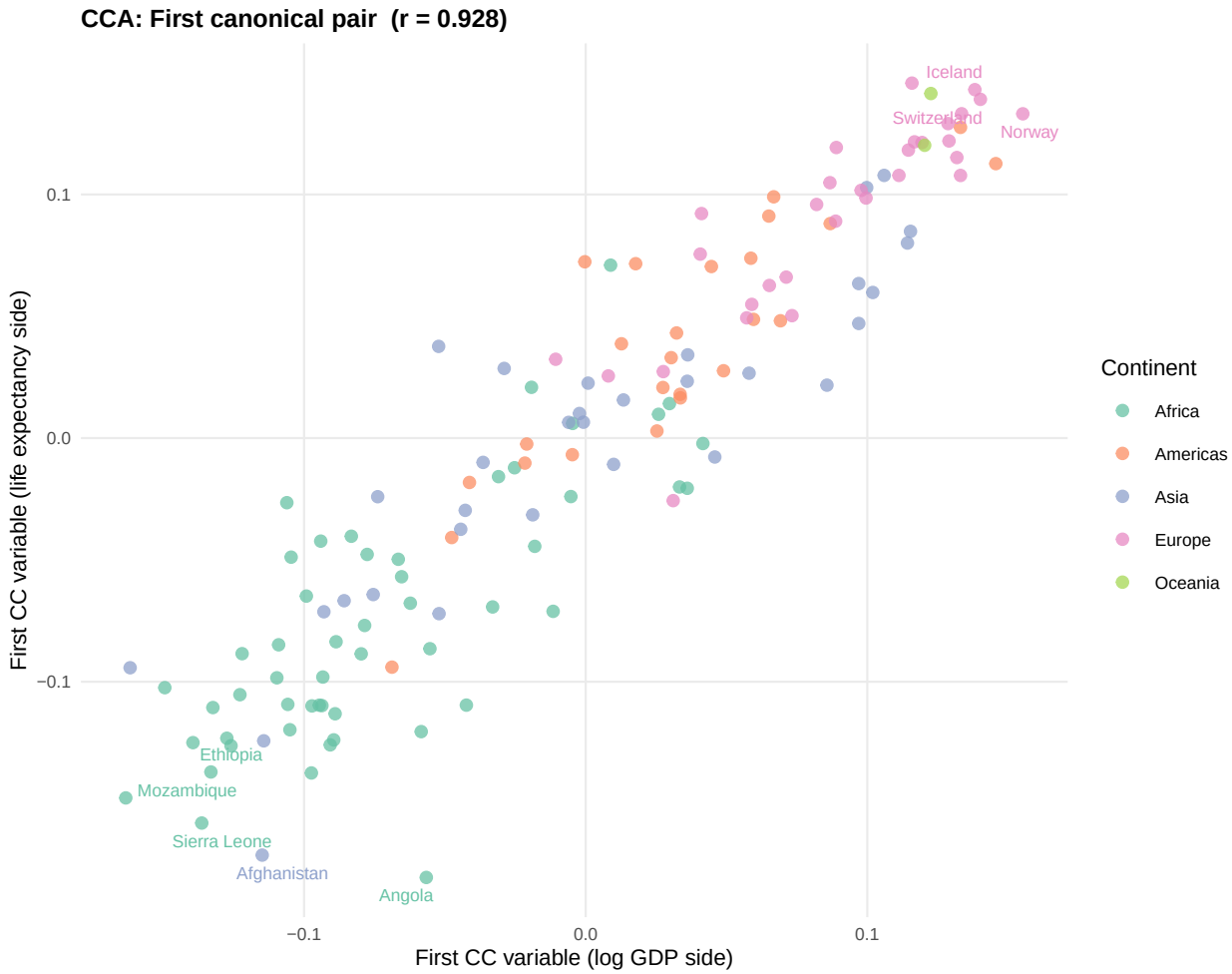

Figure 8: First pair of canonical variables, coloured by continent.

1.3.1 Canonical correlations

The first canonical correlation is 0.93, and subsequent correlations decay to 0.63, 0.54, and 0.5, with the remainder below 0.40. The first pair therefore carries most of the shared linear structure between the two blocks. Looking at CC2 directly, the GDP-side loadings start near zero in 1952 (0.003), rise to 0.11 around 1977, and fall back to −0.11 by 2007, while the life-expectancy loadings are uniformly small and negative across all 12 years (−0.05 to −0.19). The GDP pattern alone describes a faint mid-period growth contrast, but it is not matched by any equivalent temporal structure on the life-expectancy side, so the two variates of CC2 are not aligned on the same temporal feature and the pair has no joint interpretation. Higher-order pairs decay further and are similarly uninterpretable.

1.3.2 Interpretation of the first canonical pair

The first canonical axis ranks countries by a joint development level that log(GDP) and life expectancy identify together. The ordering is near-identical to PCA's on either variable alone, because the 12 year columns within each block correlate at ≈ 0.99 and CCA has little room to pick a direction meaningfully different from PCA's. The continent structure reproduces the PCA pattern, with Africa lower-left and Europe and Oceania upper-right.

1.3.3 Effect of the log transform

On raw GDP the first canonical correlation drops to 0.89 from 0.93. CCA only finds linear combinations, so curvature in the raw GDP–life expectancy relationship is uncaptured; the log transform straightens it and recovers the lost correlation.

1.4 Multidimensional Scaling

Classical (metric) MDS is applied to the combined 36-dimensional dataset formed by concatenating log(GDP), life expectancy, and log(population), using Euclidean distance as specified in the coursework. The log transforms on GDP and population bring them onto a roughly comparable scale with life expectancy.

The Pearson correlation between pairwise distances in the original 36-D space and the 2D MDS projection is 0.998, so the two retained dimensions preserve the distance geometry of the full data almost exactly.

Figure 9: Two-dimensional MDS representation, coloured by continent.

1.4.1 Development axis

Dimension 1 is a development axis. The continent centroids line up from Africa at -37 through the Americas at $+18$ and Europe at $+44$ to Oceania at $+52$, the same order PCA and CCA produced. The country ordering inside is near-identical too, since all three methods project the same dominant development signal.

1.4.2 Growth direction

Dimension 2 separates countries by trajectory rather than level, with gainers like Oman and Vietnam at one end and southern African declines (Zimbabwe, Botswana) at the other. Asia's centroid sits below the horizontal because its fast-gaining countries dominate; Africa's hovers near zero because the continent contains both improving and declining cases. This is the same contrast as PC2 in the life-expectancy PCA, now found jointly with all three variables.

1.4.3 Relationship to PCA

The agreement between the MDS and PCA axes is not a coincidence. Classical MDS with Euclidean distance is, up to sign, the same decomposition as PCA on the same centered data. Writing the

SVD of the centered data matrix as

$$X = UDV^T,$$

PCA returns the scores

$$T_{\text{PCA}} = XV = UD.$$

Classical MDS forms the Gram matrix $B = XX^T$ and takes the eigendecomposition

$$B = UD^2U^T,$$

from which the MDS coordinates are the leading columns of UD, the same matrix as T_{PCA}. The MDS scatter does not literally reproduce either PCA scatter because the input differs. Here MDS is applied to the 36-column concatenation of log(GDP), life expectancy, and log(popn), whereas the PCA sections ran the decomposition on the GDP and life-expectancy blocks separately.

1.4.4 Comparison with the earlier analyses

MDS recovers the same two axes the PCAs found, but with all three variables at once instead of one block at a time. The 0.997 distance correlation says development level and growth direction are the two dominant sources of variation in the full UN data.

2 Question 2

2.1 Linear Discriminant Analysis

We use LDA to predict continent from all 36 features (GDP, life expectancy, and population across 12 years). The data are split 70/30 into training and test sets.

A single fixed split with seed 42 gives a test accuracy of 0.643 on 42 held-out countries, and averaging over 50 random splits gives a more stable estimate of 0.616 ± 0.074. Both figures sit comfortably above the 0.20 five-class chance rate and above the 0.37 majority-class baseline set by Africa's sample proportion, so the classifier is extracting genuine structure from the features.

Table 2: Rows are true continents, columns are predicted.

	Africa	Americas	Asia	Europe	Oceania
Africa	13	0	1	0	0
Americas	2	2	3	0	1
Asia	2	2	5	1	0
Europe	0	0	1	7	1
Oceania	0	0	0	1	0

Per-class accuracy varies substantially. Africa is the best-classified continent, followed by Europe, Asia, and the Americas. The Oceanian country in the test set is misclassified as European. Asia's errors are spread across Africa, the Americas, and Europe in roughly equal proportions.

The per-class pattern follows from LDA's assumptions. The rule

$$\delta_c(x) = x^\top \Sigma^{-1} \mu_c - \tfrac{1}{2} \mu_c^\top \Sigma^{-1} \mu_c + \log \pi_c,$$

with μ_c the class mean, Σ the pooled within-class covariance, and π_c the prior, treats each class as Gaussian with shared Σ. Asia breaks the first assumption. Its GDP distribution is bimodal, so μ_{Asia} lands between the two modes and $N(\mu_{\text{Asia}}, \Sigma)$ concentrates where almost no Asian country actually sits, while overlapping the Gaussians for Africa, the Americas, and Europe. Oceania and Europe have far tighter within-class spread than Africa or Asia, so the pooled Σ fits neither. Africa is classified well despite being heterogeneous because its spread runs along a single unimodal axis the Gaussian can still represent.

2.2 Clustering

Three methods from the module are applied to the standardised GDP and life-expectancy blocks (24 features). k-means and the model-based clustering are fitted directly, and hierarchical clustering is applied via its linkage-distance dendrogram, using Ward's linkage as in the computer lab sessions. Ward merges the pair of clusters producing the smallest increase in within-cluster variance at each step. k is chosen from the dendrogram and the MVN-mixture BIC curve, with the same k then used for k-means so that all three methods can be compared on equal footing.

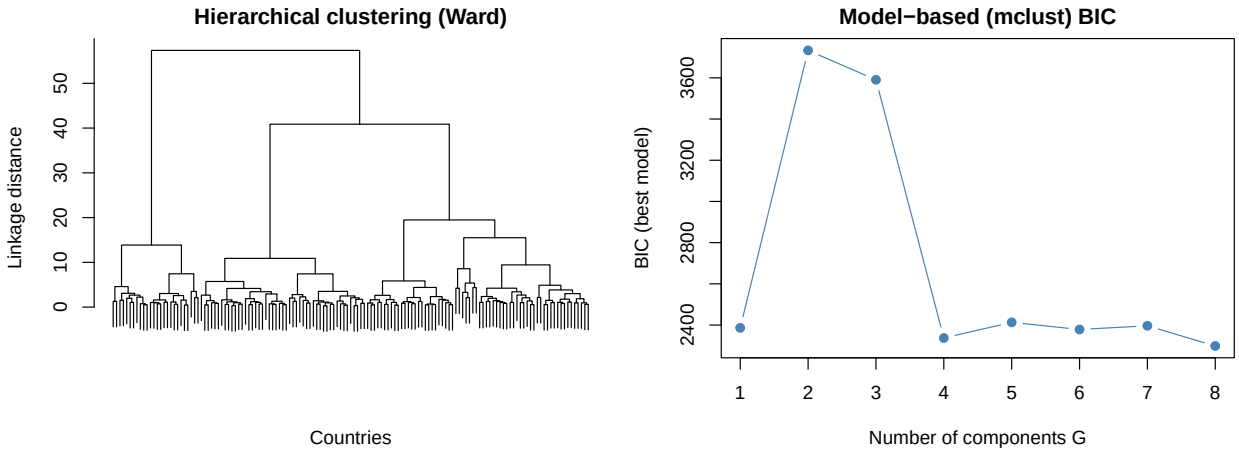

Figure 10: Hierarchical clustering dendrogram (left) and model-based BIC as a function of number of components (right).

2.2.1 Choice of k

The dendrogram's three largest merge distances are 57.4, 40.9, and 19.5, with every subsequent merge below 16. The 4-unit gap between merges 3 and 4 identifies $k = 3$ as the cleanest cut. Mclust's BIC peaks at $G = 2$ (3733), with $G = 3$ second at 3591 and $G = 4$ dropping to 2337. The 143-unit gap from $G = 2$ to $G = 3$ is small next to the 1254-unit drop to $G = 4$, so the BIC profile leaves $G = 2$ and $G = 3$ the only credible candidates and rules out the rest.

The two criteria measure different things. The dendrogram tracks where merges introduce new within-cluster spread. BIC scores per-parameter fit of a Gaussian mixture density model. PCA and MDS in Question 1 showed the data run along a single development axis, and a 2-component Gaussian fits that shape by placing one mode at each tail. A third component then costs more parameters than it gains in likelihood, so BIC prefers $G = 2$ even when the data along the axis split into three groups.

We use $k = 3$. The dendrogram has a clean three-merge cut, PCA and MDS already showed a low–middle–high ordering, and the $G = 3$ mixture is the second-best BIC fit. A two-cluster cut collapses the middle group into the low one.

2.2.2 Agreement between methods

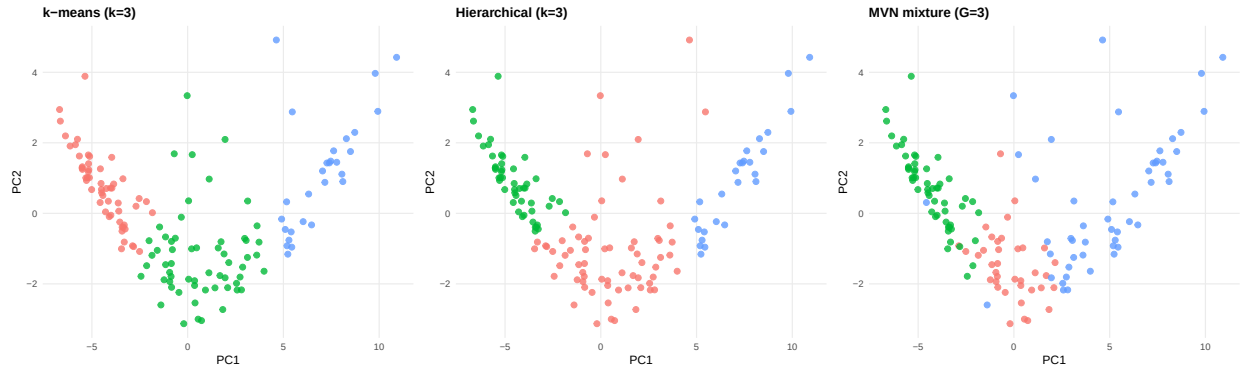

Figure 11: Cluster assignments at k = 3 for k-means, hierarchical clustering, and the MVN mixture.

At $k = 3$ the Adjusted Rand Index is 0.84 between k-means and hierarchical clustering, 0.57 between k-means and the MVN mixture, and 0.51 between hierarchical and the MVN mixture. The k-means and hierarchical results are nearly identical, consistent with both being variance-minimising methods on the same standardised features. The MVN mixture disagrees more because mclust fits full-rank covariance structures (varying shape, equal volume) rather than the spherical within-cluster geometry that k-means and Ward's method assume, and the covariance flexibility lets it draw cluster boundaries that follow the anisotropic spread of the data rather than minimising within-cluster variance.

2.2.3 Cluster interpretation

The clusters split by development level. One holds 18 of 30 European countries, 6 Asian, and both Oceanian, sitting at the high end. A second holds 42 of 52 African countries with 10 Asian and 2 American, at the low end. The third is the middle group, dominated by the Americas (21 of 25) with 16 Asian and the remaining 12 European countries. Continent is a reasonable but imperfect proxy. Africa and Europe fall almost entirely into the low and high clusters, while the Americas and Asia split across the middle and the extremes.

2.3 Linear Regression

We predict 2007 life expectancy from the 11 GDP columns for 1952–2002, excluding 2007 GDP to avoid contemporaneous leakage. The 11 predictors correlate at ≈ 0.99, so OLS alone is a poor baseline; we compare it to ridge regression and principal component regression (PCR), each fitted on raw GDP and on log(GDP).

Performance is measured by leave-one-out cross-validation. Each country is held out in turn, the model is refit on the remaining 140, and the held-out country is predicted from that fit.[2] The score is

$$\mathrm{RMSE}_{\mathrm{LOOCV}} = \sqrt{\frac{1}{141}\sum_{i=1}^{141}(y_i - \hat{y}_i)^2},$$

where each $\hat{y}_i$ comes from a fit that never saw country i.

[2]Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning* (2nd ed., 2009), Section 7.10.

Ridge and PCR have hyperparameters (α and k), which we tune by nested cross-validation. The LOOCV above is the outer loop. Within each outer fold, an inner 10-fold CV on the 140 training rows picks α or k, and the selected value is used to refit on those 140 rows and predict the held-out country. The held-out country never enters the inner loop, so it cannot influence the hyperparameter used to score it. A single flat LOOCV that chose α or k from all 141 residuals would feed each country's residual back into its own hyperparameter selection, pulling the reported RMSE below the true out-of-sample value. OLS has no hyperparameter and is unaffected.

2.3.1 Ridge regression and principal component regression

Ridge regression minimises

$$\|y - X\beta\|^2 + \alpha\|\beta\|^2,$$

adding a penalty on the squared length of β to the usual least-squares loss. When predictors are near-collinear, OLS fits tend to pair large positive coefficients with large negative ones that mostly cancel on the fitted values. The ridge penalty rules these out and returns a smaller, more stable set of coefficients in exchange for a small amount of bias. The parameter α sets the trade-off. At $\alpha = 0$ Ridge reduces to OLS, and as $\alpha \to \infty$ the coefficients are driven to zero. We search α on a 50-point log-spaced grid in $[10^{-2}, 10^3]$.

Principal component regression replaces the 11 near-collinear year columns with their first k principal components and fits OLS in that reduced space. The PC directions beyond the leading few carry almost no variance in the GDP block, so including them in OLS would inflate the fit's sensitivity to them without supplying any real signal. Cutting the number of components at k eliminates those directions outright.

2.3.2 Ridge and PCR as shrinkage on principal directions

Both estimators can be written as filters on the principal directions of the predictor matrix. Let the SVD of the centered X be

$$X = UDV^T,$$

where D holds the singular values in decreasing order. The OLS fit projects y onto the column space of X, weighting every PC direction equally, while Ridge and PCR replace those unit weights with component-wise filters,

$$\hat{y}_{\text{OLS}} = \sum_i u_i u_i^T y, \qquad \hat{y}_{\text{Ridge}} = \sum_i \frac{d_i^2}{d_i^2 + \alpha} u_i u_i^T y, \qquad \hat{y}_{\text{PCR}} = \sum_{i=1}^{k} u_i u_i^T y.$$

Ridge's factor is nearly one on high-variance directions and near zero on low-variance ones, giving a smooth shrinkage. PCR is the hard version, keeping the first k directions untouched and dropping the rest. In the GDP block the first three singular values dominate the remaining eight by orders of magnitude, so both filters produce fits built almost entirely from the first three PC directions. Ridge's nested-CV RMSE and PCR's are two smoothings of the same low-rank structure, explaining why they land so close.

On log(GDP) the selected α is stable across the 141 outer folds (range 3.6–11.5, median 7.2). On raw GDP it is less stable (range 0.01–47.1, median 23.3), reflecting a flat cross-validation surface on the raw scale where different subsets of 140 countries favour very different penalties.

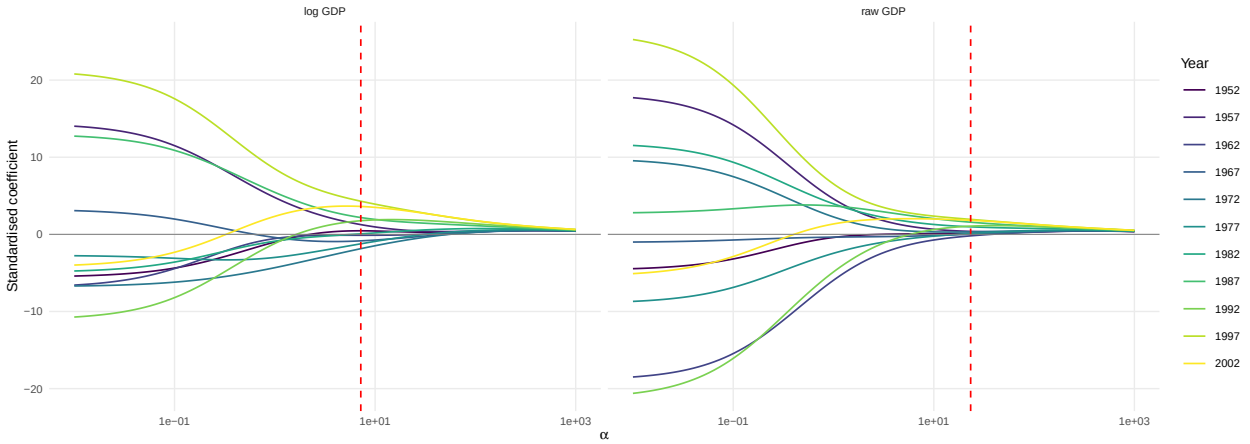

Figure 12: Standardised Ridge coefficients as a function of the penalty α, on raw GDP (left) and log GDP (right). The vertical dashed line marks the median α selected across the 141 nested-LOOCV outer folds.

At low α the eleven year coefficients are large and opposing in sign. With near-collinear predictors, OLS can pair up large coefficients that mostly cancel when predicting the response. Increasing α shrinks them toward zero and flattens the year-to-year differences. On log(GDP) the nested-CV median sits in the intermediate region, where shrinkage stabilises the coefficients without yet removing their signal.

Table 3: Nested leave-one-out cross-validated performance.

Model	LOOCV RMSE	LOOCV R2
OLS (raw GDP)	9.26	0.41
OLS (log GDP)	7.64	0.60
Ridge (raw GDP)	9.04	0.44
Ridge (log GDP)	7.25	0.64
PCR (raw GDP, k=2)	8.97	0.44
PCR (log GDP, k=3)	7.15	0.65

OLS on raw GDP gives a LOOCV RMSE of 9.26 years and $R^2 = 0.41$, while OLS on log(GDP) gives RMSE 7.64 and $R^2 = 0.60$, a one-and-a-half-year reduction in prediction error from the transform. Ridge on log(GDP) improves this further to RMSE 7.25 and $R^2 = 0.64$, while Ridge on raw GDP reaches RMSE 9.04 and $R^2 = 0.44$, a modest gain over the raw-scale OLS. GDP's skewness makes the linear relationship with life expectancy non-linear on the raw scale but close to linear on the log scale, and the regression results express the cost of ignoring this as 0.19 units of R^2.

PCR has the lowest RMSE. On log(GDP) the inner CV picked $k = 3$ components in every one of the 141 outer folds, giving nested RMSE 7.15 and $R^2 = 0.65$. The first three GDP components already explain 99% of the variance, so dropping any single country never changes which directions dominate the design. PCR on raw GDP peaks at $k = 2$ with RMSE 8.97, still short of the log-based models. The transform matters more than the multicollinearity fix.

For PCR the held-out country can't influence the k used to score it, so the nested RMSE is an

unbiased estimate of out-of-sample RMSE. For Ridge the inner-selected α varies across outer folds, so the nested score exceeds a flat LOOCV. Both outperform OLS on log(GDP) by addressing the near-collinear year columns.

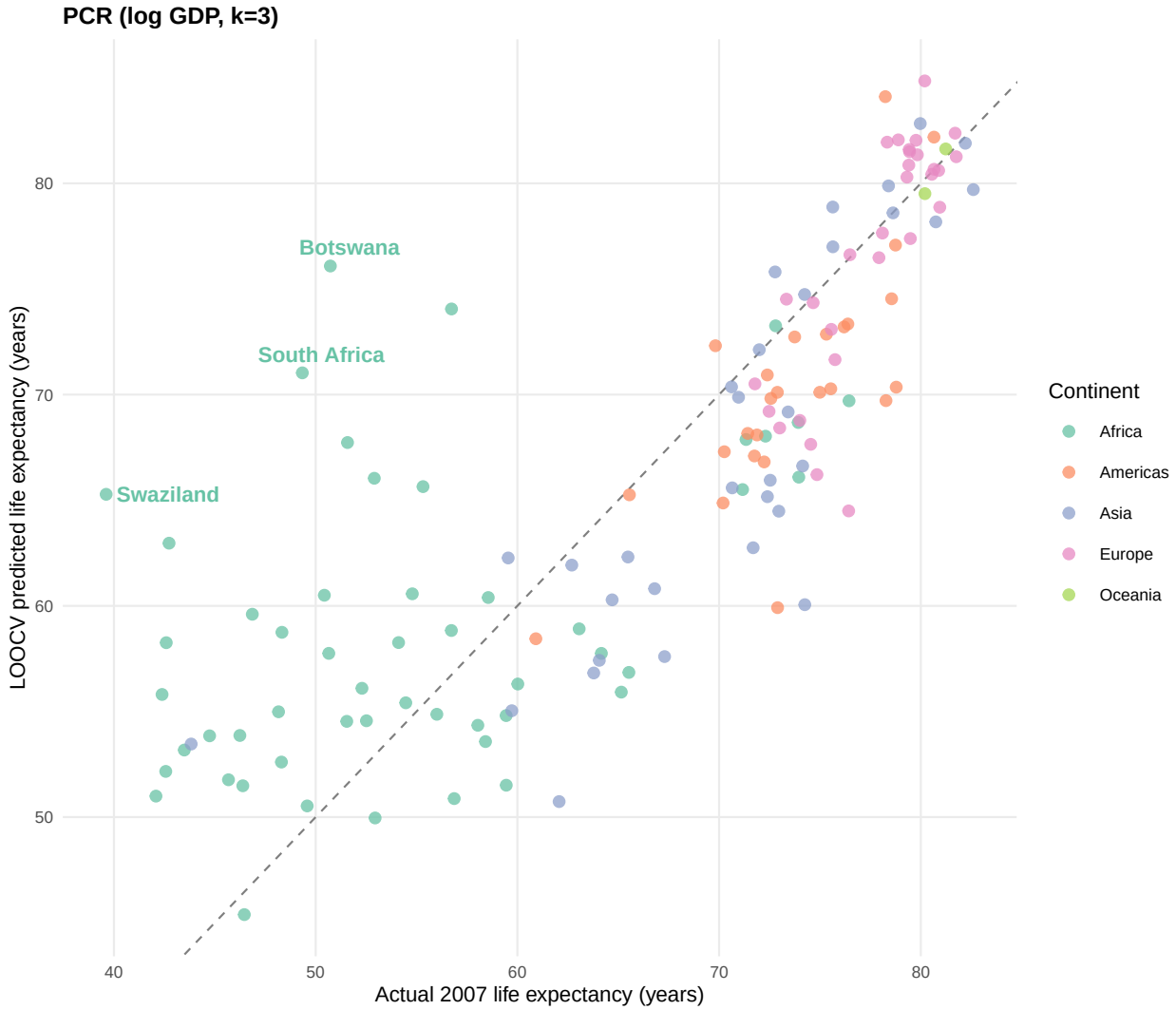

Figure 13: Leave-one-out predictions against actual 2007 life expectancy for the best model.

The 141 leave-one-out residuals have SD 7.17. All 7 countries beyond 2 SD are African and over-predicted. Swaziland, Botswana, and South Africa miss by more than 20 years, then Angola, Lesotho, Equatorial Guinea, and Gabon by between 15 and 17. Zambia, Namibia, and Nigeria sit just inside the threshold, extending the pattern.

These countries hit GDP-decoupled shocks the model can't see. The southern African group lost life expectancy from the 1990s onward without a proportional fall in income, an HIV-driven health collapse, while the oil-rent economies (Equatorial Guinea, Gabon, Angola) earned resource income concentrated in a narrow elite without matching public-health gains. The two negative-residual outliers run the opposite way, with Vietnam (-14.2) and Nicaragua (-13.0) gaining life expectancy faster than GDP. Every combination of transform, regularisation, and component count we tried left

the same countries furthest from the diagonal.